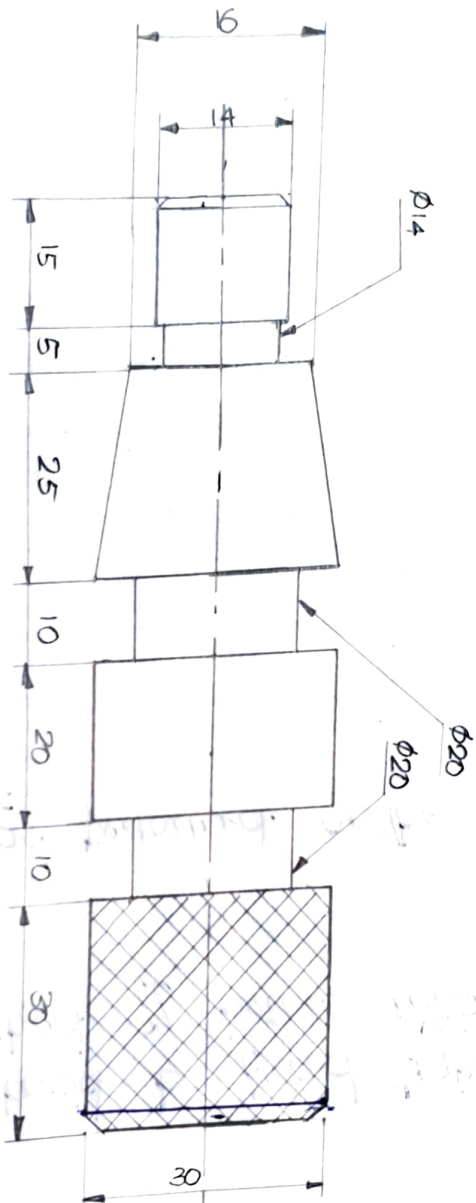


Grooving



All dimensions are in "mm"
Scale : 1 : 1

GROOVINGAim

To construct given figure into model using given MS Rod by practising Grooving.

MATERIALS REQUIRED

Work piece material : MS rod

Work piece size : 30mm dia and 15mm length.

Tools Required :-

Lathe with standard accessories.

Cutting tool

Grooving tool

Spanner.

Vernier caliper.

PRINCIPLE

Grooving operation is done to create shoulder with a small length. This is done to ensure correct fit & rotating parts when the threaded is required to run full length. & the parts to a shoulder. A groove is machined to allow full travel & nut grooving operation is also done to fit and when grooving is done all the way to center. The operation is called posting.

PROCEDURE :

- The workpiece is centered and clamped securely on the chuck.

- The tool are mounted and centered.

- The workpiece is first rotated at normal cutting speed.

- The workpiece is drilled at face and held at tail stock.

- Tool is positioned and small impression of grooving is made

- The speed is desired. To lowest speed and groove is made

- using small cross feed.

- After grooving the correct diameter the tool is pulled back machine switch off and the work piece and tool are unclamped.

PRECAUTION.

- The workpiece and tool must be securely clamped.
- The workpiece must be supported by 2 centers.
- Use required feed and speed only.

RESULT

Grooving operation was successfully completed to produce the required product successfully.